Advanced Labs I　(AY2025)

Theme			
Optoelectronics			
Dates of Experiments	13/01/2026, 26/01/2026, 2/01/2026	Report Submission Date	17/02/2026
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Date of Receipt	
Recipient	

By the theme instructor,

Date of return	
Rating /10	
Are the principles, equipment, and experimental procedures fully understood? a b c d e	
Are the experimental data properly analyzed? a b c d e	
Are the experimental results presented appropriately? a b c d e	
Is the discussion substantial? a b c d e	
Is the structure of objectives, methods, results, discussion, etc. appropriate? a b c d e	
Is it carefully described? a b c d e	
Comments	

Optoelectronics

Purpose:

to understand the basic operation principles of LEDs, LDs, and Solar Cells through the observation of their emission spectra, photoresponse characteristics, quantum efficiencies and classifying them into different types based on known information. For LEDs, emission graphs are created and categorized for each type of LED. For solar cells, emission graphs and J-V graphs are created and categorized for each solar cell.

Report Assignments (Part 1: LED, Part 2: Xe lamp and solar cells)

1-1 Make graphs of each emission spectrum (wavelength [nm] versus emission intensity [W]) for red-light LED (at RT), blue-light LED (at low temperature and 60°C), yellow-light LED (at RT), and white-light LED (at RT).

Each graph of the emission spectrum is calculated and created using photointensity (W) as the vertical axis and wavelength (nm) as the horizotal axis. To obtain the photointensity values, we first divide the experimental light intensity voltage values by the resistance value 9000 ohms to get the current values. By dividing the current values (A) by the spectral intensity specific to each wavelength in Table 1 in the textbook (A/W), we can get the spectral intensity photointensity (W).

The graph of emission spectrum for red LED is in Figure 1.

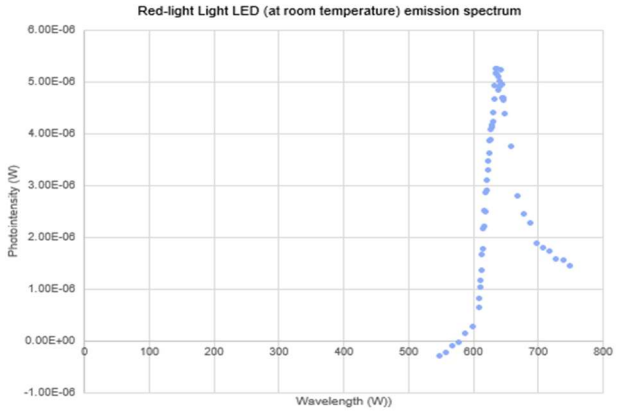


Figure 1 red-light LED emission spetrum graph at room temperature

The graphs of emission spectrum for for the blue-light LED at −5℃ and 60℃ are shown in Figures 2 and 3 respectively.

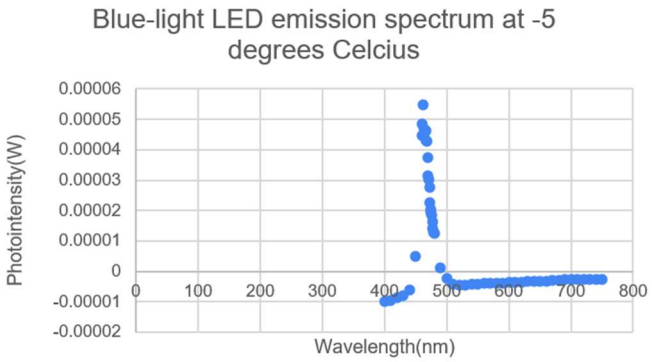


Figure 2 Blue-light LED emission spectrum at -5 degrees Celsius

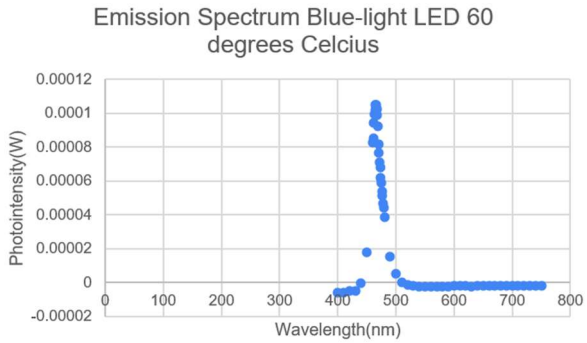


Figure 3 Blue-light emission spectrum at 60 degrees Celsius

The graph of emission spectrum for yellow light LED at room temperature is in Figure 4.

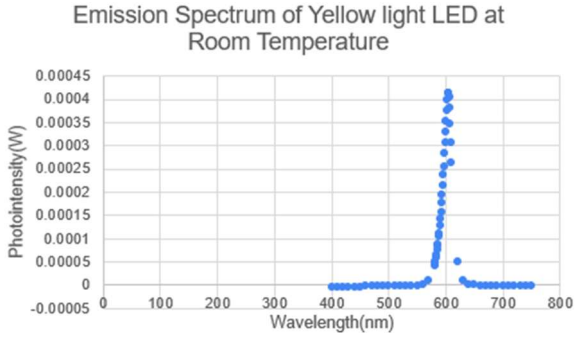


Figure 4 Yellow-light emission spectrum graph at room temperature

The graph of emission spectrum for white light LED at room temperature is in Figure 5.

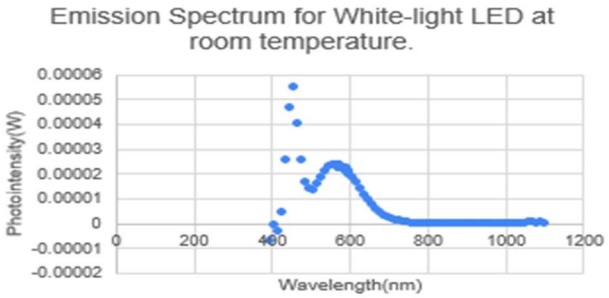


Figure 5 Yellow-light LED emission spectrum at room temperature graph

1-2 Obtain a peak wavelength [nm] in each spectrum and its energy [eV] for a red-light LED (at RT), a blue-light LED (at low temperature and 60°C), a yellow-light LED (at RT) , and a white-light LED (at RT)

To find the peak wavelengths , we first identify the highest points on the graph, except for blue light which is found by identifying the highest point and averaging it with the point at which the emission intensity becomes half. The white light-LED has 2 peaks. The peak wavelengths and ttheir respective bandgap energies (eV) are shown in the table from Excel in Figure 6.

LED	Temperature	Half-Maximum Wavelengths [nm]		Peak Wavelength [nm]	Photon Energy [eV]
Red	RT			644	1.925
Yellow	RT			605	2.050
Blue	-5°C	444	472	458	2.707
	60°C	455	478	466.5	2.658
White	RT			460	2.696
				580	2.138

Figure 6 Table of peak wavelengths of each colour-LED and their respective photon energies

The band-gap energy (E) of each light color at their respective wavelengths can be determined by the following formula,

$$E = \frac{\hbar c}{\lambda}$$

where $\hbar$ is the Planck's constant, c is the speed of light, and $\hbar$ is its peak wavelength. For calculations, it was assumed that $\hbar c$ =1240eV×nm.

1-3 There are several types for white-light LEDs. From the number of peak wavelength you got for a white-light LED in this study, discuss how the white-LED emits "white light".

White light is a combination of different wavelengths of light that appear white when emitted altogether. This can be seen in the table in Figure 6 that the white light made

by mixing the lights from Red, Blue, and Yellow LEDs. Dual-peak wavelengths (Hybrid phosphor + LED combinations) are LEDs mix two distinct wavelengths, in this case blue light from an LED and red light from a phosphor, to create white light. Another example of dual-peak wavelengths (Hybrid phosphor +LED combinations) is a white LED is made of a particular type of blue LED covered with a YAG:Ce phosphor. The white color is produced by a mixture of blue and yellow colors.

1-4 Explain the reason why you need to change the filter when you measure the emission spectrum of a white-light LED? Weswitch the filter when measuring the White-light?

We switch the filter when measuring LED's emission spectrum because:

1. The straight white light from the LED includes other wavelengths besides our target wavelength. These extra wavelengths can pass through the diffraction grating without any trouble. So, to get rid of these unwanted wavelengths, we use a special filter (like the R-64 filter in our experiment) that blocks them. It's worth mentioning that this problem mainly occurs when we measure wavelengths higher than about 690 nm.
2. To control intense peaks and prevent dominant wavelengths (e.g., blue light or red light) from saturating the detector.
3. To improve resolution by isolating specific parts of the spectrum such as low or high frequencies

1-5 Investigate why the emission peak intensity of a blue-light LED and its wavelength changed when the temperature of the LED was changed?

The light output power decreases as the temperature (T) increases. This can be seen in the table in Figure 6, where the band-gap energy (photon energy) for blue-light LED at low temperature is higher (2.707 eV) than that of the blue-light LED at 60℃ (2.658 eV). The reason is because: (1) within the quantum wells that emit light, more injected electrons recombine non-radiatively to emit heat rather than light at high T
(2) at high T, more electrons leak out of the light generating quantum well regions.
At the Supplementary Material 1 in the textbook, while the rate of radiative recombination (the process that yields a photon) stays the same when the temperature goes up, the rate of non-radiative recombination (which leads to the excitation of another electron instead of photon emission) sees an increase. The radiative recombination ratio, R_{rad}, is directly proportional to the light output of the LED. Thus, as the temperature rises, R_{rad} diminishes, and since it's directly proportional to light output, the LED's emission intensity correspondingly declines with an increase in temperature.

1-6 The emission spectrum of a red-light LD became pretty sharp at I > Ith compared to that at I < Ith. Investigate the device structure of LDs and discuss why this happens.

We did not measure red LDs in our experiment, only LEDs. Thus, we will discuss about the red LED. When we significantly increase the current flowing into the Red-light LED, the light intensity in the area where the light is produced goes above the threshold level. This means there are a lot more carriers and electrons ready to come together and create light particles, or photons. As a result, the LED starts to emit a lot more light.

Figure 7 shows the spontaneous emission spectrum of red light with a changing input current (I).

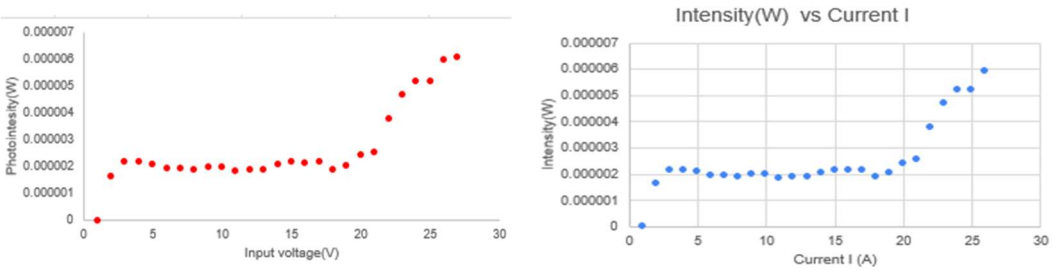


Figure 7 Spontaneous Emission of Red Light. Left graph : Intensity (W) to input voltage V (volts). Right graph: Intensity(W) to current derived from output voltage (current I= output voltage / resistor).

It can be seen, especially in the right graph of figure 7, that the emission spectrum of a red-light LED became pretty sharp at I > Ith compared to that at I < Ith. because the device transitions from spontaneous emission to stimulated emission, behaving like a laser diode (LD). At I > Ith the carrier density reaches a critical point where there are more electrons in the conduction band than in the valence band which is known as population inversion. This creates the necessary condition for stimulated emission to dominate.

Part 2: Solar cells

We used three solar cells for our experiments. The three solar cells are solar cell A, solar cell B, solar cell C. We will do experiments to determine which category each solar cell belongs to: amorphous Si or crystalline Si, which have a single pn junction; or GaAs-based compound solar cell, which is a tandem (stacked two pn-junction diodes with different bandgap energies) solar cell. This will be explained in question 2-3.

Solar cells A, B, and C are in Figures 8,9, and 10, respectively.

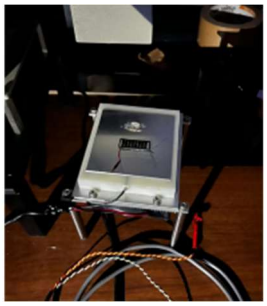


Figure 8 Solar cell A



Figure 9 Solar cell B

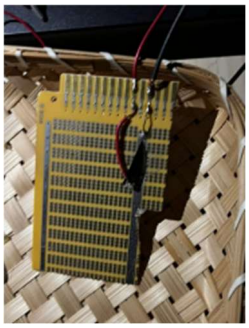


Figure 10 Solar cell C

2-1 Make a graph of the emission spectrum of the Xe lamp.

The emission spectrum of the Xe lamp is in Figure 11. The absolute values of the output current, which is measured in μ W, are divided by the area 0.5 m^2 × respective photoresistivity values at each wavelength, to obtain the photointensity values (W) in the graph.

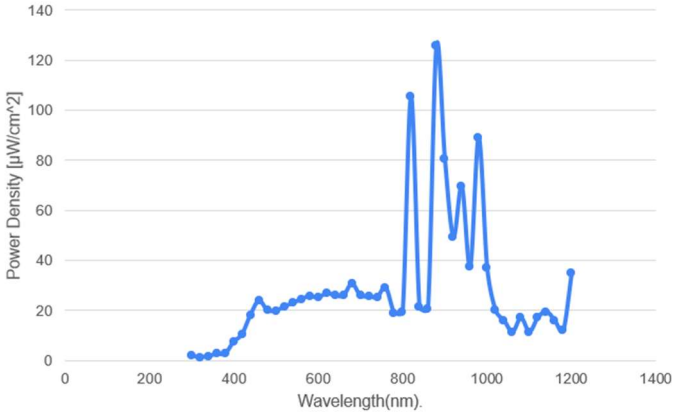


Figure 11 Emission Spectrum of the Xe Lamp

2-2 From the photo-response spectra of solar cells A and B in Experiment 2, make graphs showing their quantum efficiency spectra, wherein the vertical axis is quantum efficiency η [%] and the horizontal axis is λ wavelength [nm].

The original photo-response spectra of solar cells A, B, and C are in figures 12,13,14, respectively.

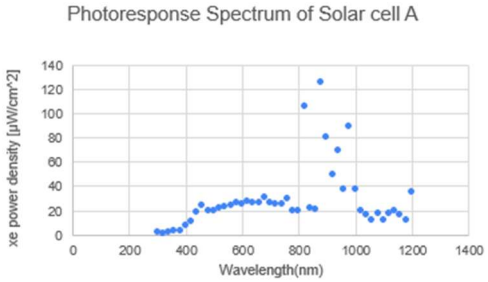


Figure 12 Photoresponse spectrum of solar cell A

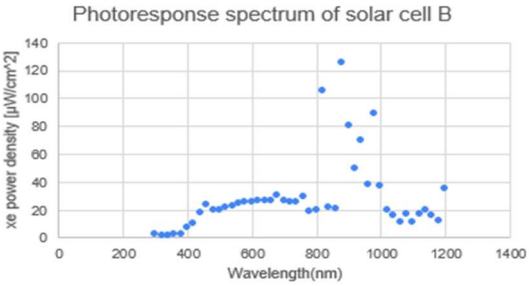


Figure 13 Photoresponse spectrum of solar cell B

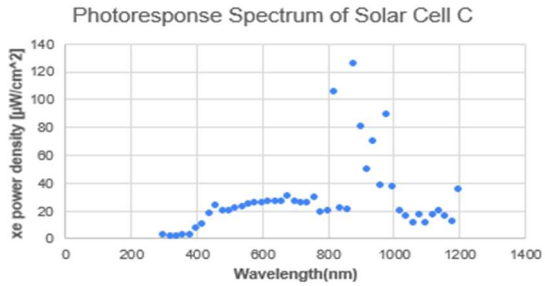


Figure 14 Photoresponse spectrum of solar cell C

The quantum efficiency represents the percentage of incident photons that generate electron-hole pairs or the conversion efficiency of photons into electrons. For each wavelength, the quantum efficiency can be derived from the photo-response spectra shown in figures 12,13, and 14 using the following formulas:

$\boldsymbol{\eta = \Phi_{photo} \ / I_e}$

where $\boldsymbol{\Phi_{photon}}$ is the photon emission rate (per second) and Ie is the injected current calculated by $\boldsymbol{\Phi_{photon}}$= P/E where P is the emission intensity [W/cm2] and E [eV] is the energy of the photon. And, Ie = I/q where I [A] is the supplied current and q [C] is the elementary charge. The quantum efficiency was then plotted against the wavelength to create the quantum efficiency graphs shown below.

The quantum efficiency spectra of solar cells A, B, and C are in figures 15,16, and 17 respectively.

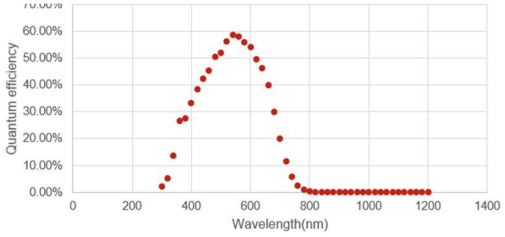


Figure 15 The quantum efficiency graph from photoresponse spectrum of solar cell A

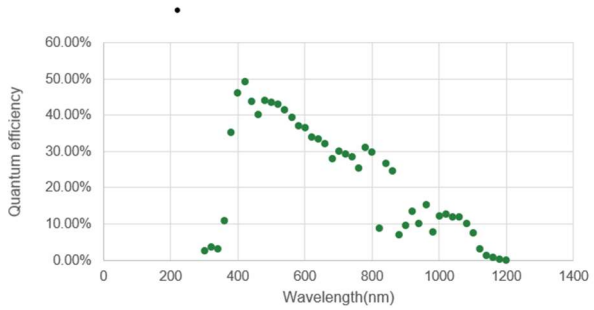


Figure 16 The quantum efficiency graph from photoresponse spectrum of solar cell B

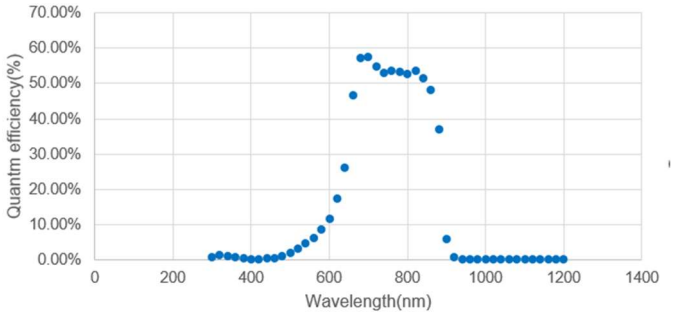


Figure 17 The quantum efficiency graph from photoresponse spectrum of solar cell C

2-3

Explain which solar cell is which

From the results of our experiments, we've determined the types of solar cells:

Solar Cell A is made of Amorphous Silicon (a-Si), Solar Cell B is a Tandem Structure with Gallium Arsenide (GaAs) compounds), and Solar Cell C is made of Crystalline Silicon (c-Si. We concluded this from their Photoresponse spectra and Quantum Efficiency (QE) spectra.

1. Solar Cell A showed one peak followed by a gradual drop in QE, indicating ongoing recombination, typical for amorphous silicon. Based on the known characteristics of a-Si cells, Amorphous silicon cells absorb strongly in the visible and UV range (300–750 nm) but show poor response in the near-infrared region due to a wider band-gap (1.7 eV). Referring to the graphs, the photo-response spectrum of solar cell A shows a peak around the visible and UV range and steeply declines in the IR range
2. Solar Cell B displayed two peaks in its QE spectrum, aligning with the expected behavior of tandem-structure cells.
3. Solar C had an initial peak and then a sharp fall in QE, stabilizing afterward, which is characteristic of crystalline silicon.

2-4 The relationship between quantum efficiency η [%] and photo-responsivity R [A/W] is given by η =R× $\hbar\omega$ ×100 [%]. Derive this relationship.

The quantum efficiency $\boldsymbol{\eta} = \frac{I_p \hbar \nu}{q P_{in}}$

Where Ip is the photocurrent, $\hbar\nu$ is the photon energy obtained from the Planck's constant $\hbar$ and photo frequency ν, q is the elementary charge and Pin is the incident optical power

$$R = \frac{I_p}{P_{in}}$$

Thus $\boldsymbol{\eta} = R\frac{\hbar}{q}$ = R $\hbar\omega$

To convert to percentage format： η =R× $\hbar\omega$ ×100 [%]

2-5 Make graphs of J-V and JxV-V characteristics for solar cells A, B, and C, and obtain the value of load resistance R which makes the output power become the maximum.

A. Solar cell A

The current density J-voltage V and power density JxV-voltage V characteristics for solar cells A, respectively, in Figures 18,19, 20,21. Figures 18 (J-V graph) and 19 (JxV-V graph) are when the solar cell at 25°C. (room temperature).

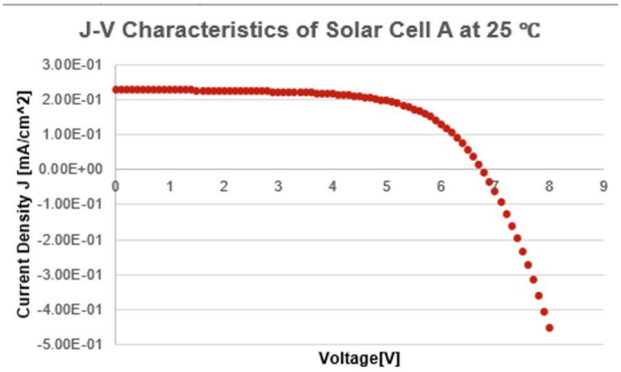


Figure 18 J-V characteristics of solar cell A at 25 degrees Celsius

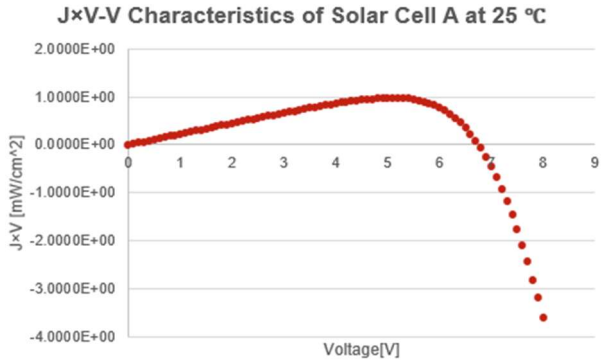


Figure 19 J X V-V characteristics of solar cell A at 25 degrees Celsius

For solar cell A at 25℃, the value of load resistance R which makes the output power become the maximum (the maximum power density J x V is 0.98073 $\left[\frac{mW}{cm^2}\right]$) is

$$R_{max} = \frac{(voltage\ at\ where\ J \times V\ is\ the\ maximum)}{[current\ density\ \ J\ \ value\ at\ where\ JxV\ is\ the\ maximum] \times 10^{-3} \times area\ of\ solar\ cell\ A}$$

$$= \frac{5.1\ V}{(1.92\times10^{-1}\left[\frac{mA}{cm^2}\right]) \times \frac{10^{-3}}{mA} A \times 3\ (cm^2)} = 8840.35\boldsymbol{\Omega}$$

Figures 20(J-V graph) and 21 (JXV-V graph) are when solar cell A is at 60°C. Even though we meant to measure in 60°C, the actual data recorded became 50°C.

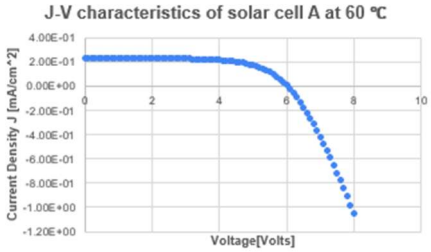


Figure 20 J-V characteristics of solar cell A at 60 degrees Celsius

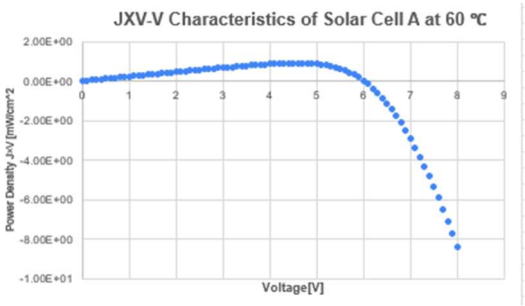


Figure 21 JxV-V characteristics of solar cell A at 60 degrees Celsius

For solar cell A at 60℃, the value of load resistance R which makes the output power

become the maximum (the maximum power density J x V is 0.910 $\left[\frac{mW}{cm^2}\right]$) is

$$R_{max} = \frac{(voltage\ at\ where\ J \times V\ is\ the\ maximum)}{[current\ density\ J\ \ value\ at\ where\ JxV\ is\ the\ maximum] \times 10^{-3} \times area\ of\ solar\ cell\ A}$$

$$= \frac{4.6\ V}{(1.98\times10^{-1}\left[\frac{mA}{cm^2}\right]) \times \frac{10^{-3}}{mA} A \times 3\ (cm^2)} = 7.75 \times 10^3 \boldsymbol{\Omega}$$

As can be seen above for solar cell A, the maximum JxV value and its corresponding J value is greater at higher temperature 60℃ . The maximum load resistance R_{max} value is smaller at higher temperature 60℃ compared to the respective values at room temperature 25℃.

B. Solar cell B

Figures 22 and 23 show the J-V and JxV-V characteristics for solar cells B, respectively. Figures 22 and 23 are when the solar cell B is at 25°C. (room temperature)

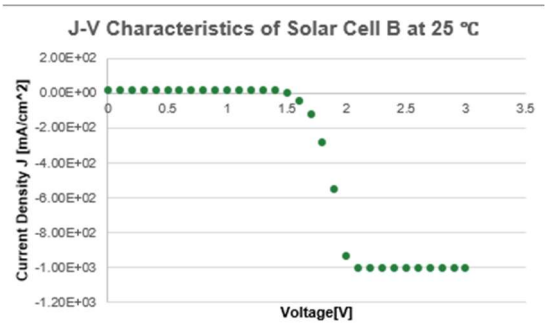


Figure 22 J-V characteristics of Solar Cell B at 25 degrees Celsius

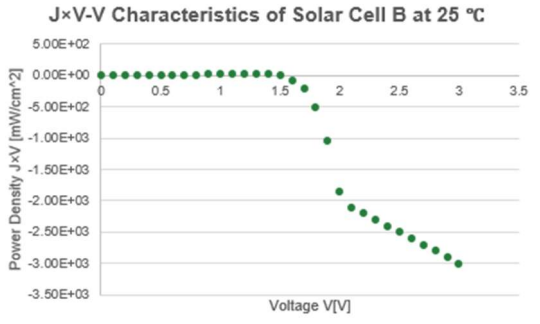


Figure 23 JXV-V characteristics of solar cell B at 25 degrees Celcius

For solar cell B at 25°C, the value of load resistance R which makes the output power

become the maximum (the maximum power density J x V is 23.3 $\left[\frac{mW}{cm^2}\right]$) is

$$R_{max} = \frac{(voltage\ at\ where\ J \times V\ is\ the\ maximum)}{[current\ density\ J\ \ value\ at\ where\ JxV\ is\ the\ maximum] \times 10^{-3} \times area\ of\ solar\ cell\ B}$$

$$= \frac{1.3\ V}{(1.80\times10^{1}\left[\frac{mA}{cm^2}\right]) \times \frac{10^{-3}}{mA} A \times 1\ (cm^2)} = 72.4233983\boldsymbol{\Omega}$$

C. Solar cell C

Figures 24 and 25 show the current density J-voltage V and power density JxV-voltage V characteristics graphs for solar cells A, respectively. Figures 20 and 21 are when the solar cell at 25°C. (room temperature)

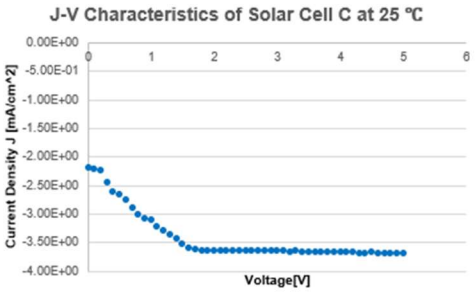


Figure 24 J-V characteristics graph of solar cell C at 25 degrees Celcius

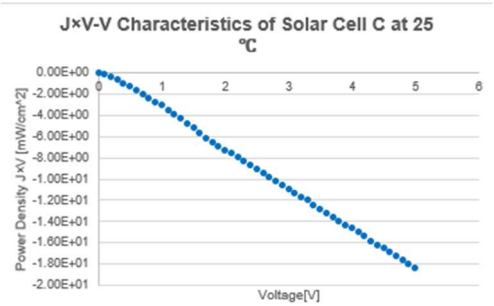


Figure 25 JxV-V characteristics graph of solar cell C at 25 degrees Celsius

For solar cell C at 25°C, every J xV value is negative except for $0\frac{mW}{cm^2}$. Thus the maximum power density is $0\frac{mW}{cm^2}$ at voltage 0 V and current density J -2.206$\left[\frac{mA}{cm^2}\right]$.

The value of load resistance R which makes the output power become the maximum (the maximum power density J x V is 0 $\left[\frac{mW}{cm^2}\right]$) is

$$R_{max} = \frac{(voltage\ at\ where\ J \times V\ is\ the\ maximum)}{[current\ density\ J\ \ value\ at\ where\ JxV\ is\ the\ maximum] \times 10^{-3} \times area\ of\ solar\ cell\ B}$$

$$= \frac{0\ V}{(-2.206\ [\frac{mA}{cm^2}]) \times \frac{10^{-3}}{mA} A \times 1\ (cm^2)} = 0\boldsymbol{\Omega}$$

2-6 Regarding solar cell A, make a graph, in which you can compare the J-V characteristics under illumination measured at RT and at 60℃

The graph showing comparison of J-V characteristics at room temperature 25℃ and 60℃ is shown in Figure 26.

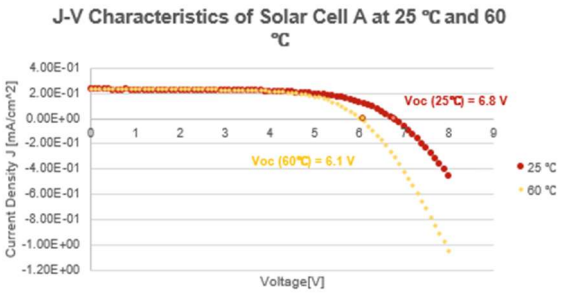


Figure 26 J-V characteristics of Solar Cell A at different temperatures

The current density J values of solar cell A are initially similar at both temperatres. Then somewhere between 5-6 volts, the J values decrease more at 60℃ compared to 25℃.

Relating to question 2-7, the Voc value at 60℃ is less than℃.

2-7 According to the graph you made above, the Voc decreased when the temperature of the solar cell increased. Explain the reason why the Voc decreases at elevated temperatures.

An equation for open-circuit voltage V_{oc} is found by setting the net current equal to zero in the solar cell equation to give

$$V_{oc} = \frac{nkT}{q} \ln\left(\frac{I_L}{I_0} + 1\right) \qquad (1)$$

V_{oc} decreases with temperature because I_0 increases rapidly with temperature primarily due to changes in the intrinsic concentration n_i.

Increases in temperature reduce the bandgap of a semiconductor, thereby effecting most of the semiconductor material parameters. The decrease in the band gap of a semiconductor with increasing temperature can be viewed as increasing the energy of the electrons in the material. Lower energy is therefore needed to break the bond. In the bond model of a semiconductor bandgap, a reduction in the bond energy also reduces the bandgap. Therefore increasing the temperature reduces the bandgap.
In a solar cell, the parameter most affected by an increase in temperature is the open-circuit voltage. The open-circuit voltage decreases with temperature because of the temperature dependence of I_0.

The equation for I_0 is

$$I_0 = qA\frac{Dn_i^2}{LN_D} \qquad (2)$$

where
q is the electronic charge; A is the area;
D is the diffusivity of the minority carrier given for silicon as a function of doping in L is the minority carrier diffusion length;
N_D is the doping; and n_i is the intrinsic carrier concentration given for silicon

In the above equation, many of the parameters have some temperature dependence, but the most significant effect is due to the intrinsic carrier concentration, ni. The intrinsic carrier concentration depends on the bandgap energy (with lower bandgaps giving a higher intrinsic carrier concentration), and on the energy which the carriers have (with higher temperatures giving higher intrinsic carrier concentrations). As the bandgap energy decreases with increasing temperature, the intrinsic carrier concentration increase. Increasing n_i results in higher I_0. And with higher I_0, according to the first equation (1), the V_{oc} decreases.

References for question 2-7:
[1] https://www.pveducation.org/pvcdrom/solar-cell-operation/open-circuit-voltage

[2] https://www.pveducation.org/pvcdrom/solar-cell-operation/effect-of-temperature